

UJJWAL

(410) 227-8791 · San Jose, CA 95131 · Open to Relocate · ujjwaltech.05@gmail.com

LinkedIn: /in/ujjwal5 · **Github:** [ujjwalbb30](https://github.com/ujjwalbb30) · **Portfolio:** ujjwalbb30.github.io/ujjwal5.github.io/

PROFESSIONAL SUMMARY

Detail-oriented Data Scientist with multidisciplinary background spanning data science, AI and mechanical design. 4+ years building scalable machine learning pipelines, real-time 3D perception and localization systems. Proficient in Python, with a strong background in data visualization, predictive modeling, and root cause analysis. Skilled in effectively communicating complex technical insights to stakeholders. Passionate about building embodied AI products and systems.

WORK EXPERIENCE

Data Scientist, Enfuse.io Inc, Baltimore, Maryland

Jul 2025 - Present

Architect and deploy scalable data science solutions leveraging Enfuse.io's partner products. Own and manage full product lifecycle, from ideation through deployment, ensuring seamless delivery and long-term maintenance for clients.

- **Warehouse Management System (WMS) @ AimsPlus Software**

Engineered backend APIs supporting order ingestion, planning, allocation, picking and shipping operations for a multi-site WMS handling 20K+ orders across 4+ locations daily. Developed a high-volume Package Planning module and integrated Paccurate for cubing optimization, cartonization and palletization (single, multistage). Streamlined material pickup scheduling and automated conveyor application for rate shopping, domestic and international label generation.

Data Scientist, Atos zData Inc., Baltimore, Maryland

Jun 2022 - Jun 2025

Implement modern software development principles, cutting-edge data science tools, and cloud platforms to deliver data-driven production-ready systems for clients.

- **Real-Time 3D Perception & Localization Module for Mobile Platforms**

Developed a real-time 3D perception module deployable on mobile platforms (car, robot, wearable) using Jetson Orin, enabling environment mapping and 6-DoF localization within a global map, leveraging 3D Gaussians (nerfstudio), point cloud representations and advanced camera pose regression techniques (6DGS). Utilized YOLO and SAM to perform instance segmentation and place the objects of interest in 3D environment.

- **Path Navigation for Indoor Scenario in Augmented Reality**

Engineered heuristic path planning solution for indoor AR navigation by training point cloud semantic segmentation model (RandLA-net) to automate store layout generation and pathfinding between locations (neural A*). Achieved 70% segmentation accuracy on key structures (floor, wall, ceiling) using 3000+ samples, cutting manual mapping effort by 2+ hours/store.

- **Retail Aisle Product Localization**

Built a high accuracy (90%) aisle-shelf-slot identification system for retail aisles to localize products in store and monitor inventory. Developed intuitive Gradio tools for ground truth annotation & performance analysis. Implemented unit tests ensuring product localization pipeline robustness.

Mechanical Design Engineer, Center for Research & Innovation, Havells, India

Jun 2018 - Jun 2019

- Pioneered upgradation of engineering designs & drawings of pump components per ISO GD&T practices; CAD modeled 150 components used across 20 SKUs, collaborated with cross-functional teams (30+ personnel) for prototyping & production.

Visiting Researcher, Advanced Vehicle Engineering Center, Cranfield University, U.K.

Feb 2017 - Jul 2017

- Conceptualized fabrication of Hardware-In-Loop test rig for Electric Vehicles thermal management system. CAD designed entire test bench and simulation chamber and finalized Purchase Order Request.

TECHNICAL SKILLS

Languages: Python, SQL, MATLAB, C++, Bash/Shell scripting (Linux/Unix)

Libraries: pandas, NumPy, SciPy, TensorFlow, PyTorch, Keras, scikit-learn, PySpark, Plotly, SQLAlchemy, FastAPI

Tools: Microsoft Azure (ML & Pipelines/CI-CD), Tableau, Streamlit, GitLab, Docker, VSCode, DBeaver, Insomnia

Data Science: Data Modelling, Visualization, Exploratory Data Analysis, Pattern Recognition, Statistical Modeling, Hypothesis Testing, A/B testing, Time-Series, Deep Learning, MLOps

EDUCATION

University of Maryland, Baltimore County (UMBC), Maryland

Dec 2022

Master of Professional Studies in Data Science

GPA: **3.971/4.00**

Data Analysis & Machine Learning, Natural Language Processing, Artificial Intelligence, Data Management, Ethical & Legal Issues in Data Science, Platforms Big Data Processing

Punjab Engineering College (PEC), India

May 2018

Bachelor of Technology, Mechanical Engineering

GPA: **7.19/10.00**

PATENTS

- “Pump Set Motor Assembly for Preventing Contaminant Ingress”, Indian Patent 201911005525, Aug 14, 2020